

CarbonShift — what our project actually is

A five-minute read before the demo. Read this with the workflow diagram open next to you.

In one sentence: we built a tool that makes cloud computer jobs run at the hour of the day when electricity is cleanest — so they cause less pollution, without changing the job itself at all.

The problem

Electricity is **not equally clean all day**.

When it is windy and sunny, a lot of the country's power comes from wind and solar. That electricity is clean. When the wind drops, power companies burn coal and gas to fill the gap. That electricity is dirty.

So the **same job, doing exactly the same work, causes different amounts of pollution depending on what time it runs**.

On the German grid we measured a real day: the dirtiest hour was **more than twice as dirty** as the cleanest hour. Same country, same day, just a different hour.

Why nobody does anything about it

A company sets its database backup to run at 8pm. So it runs at 8pm. Every single night. It has no idea what the power stations are burning at 8pm, and it never asks.

But here is the thing — **nobody is waiting for that backup**. It could run at 3am. It could run at 9am the next morning. Nobody would notice or care.

That is a free saving that everybody is walking past.

What we built

CarbonShift. You give it two things:

1. what the job is
2. the time by which it must be finished

It then finds the cleanest hour before that deadline and runs the job at that hour instead.

Same job. Same result. Only the timing changed. In our real test that meant **about half the CO2**.

Which jobs — this part matters

It only works on jobs that **nobody is sitting and waiting for**.

YES — safe to move

- Nightly backups
- Moving and cleaning up data overnight
- Generating monthly reports
- Training an AI model
- Deleting old files, tidying up

NO — must never be moved

- Someone logging in
- A payment going through
- An emergency alert
- A health check making sure a website is still alive

If a human being is waiting for it, delaying it is simply the wrong thing to do — and **our tool refuses to touch those jobs**. That refusal is deliberate, and it is one of the best things about the project.

How it works — the three steps in the diagram

1 DECIDE

It looks up a forecast of how clean the electricity will be for the next 24 hours. It throws away every hour that is after your deadline. Out of the hours that are left, it picks the cleanest one.

2 BOOK

It books the job with Amazon's cloud for that exact minute — like setting an alarm clock that Amazon holds, not us. From this moment our laptops can be switched off completely. The job will still run.

3 RUN & PROVE

Hours later, Amazon starts the job by itself and writes a log proving it ran. We keep a record of every decision, so afterwards you can see a dashboard, a history of everything saved, and a printable report.

What makes it good (say these if a judge asks)

It knows when <i>not</i> to act	If no hour before the deadline is cleaner than right now, it refuses to move the job. Delaying it would make things worse, so it does not.
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It does not fake the numbers	If a job cannot be proven to have run, the report leaves it out of the total and says so. It would rather claim less than claim something it cannot prove.
It really runs	This is not a simulation. Real jobs are booked in real Amazon infrastructure and really run, hours later, on their own.
It is properly tested	162 automated tests, all passing, in under two seconds.
It is free and open	Open source under the MIT licence. Anyone can download it and use it for nothing.

The five numbers to remember

49.5%	CO2 saved in our real test run — the headline number
2×	how much dirtier the worst hour of the day is than the best
251–512	the range we measured in one day, in gCO2/kWh (see glossary)
16 hours	how long the job was delayed to get that saving
162	automated tests, all passing

If you only remember one thing: the dirtiest hour of the day is twice as dirty as the cleanest one. We move jobs that nobody is waiting for into the clean hour. That is the whole idea.

Words you will hear tomorrow

Do not panic if a judge uses these. Here is what they mean.

The grid	The country's electricity supply — all the power stations together.
Carbon intensity	How dirty the electricity is at this moment. A low number means clean.
gCO2/kWh	The unit for that. Grams of CO2 released per unit of electricity used.

A cloud job	A task that a computer somewhere on the internet runs for you.
AWS	Amazon Web Services. Amazon's cloud. We rent their computers to run the jobs.
EventBridge Scheduler	Amazon's alarm clock. We tell it "run this at 9am" and it does, on its own.
Fargate	The Amazon service that actually runs our job when the alarm goes off.
CloudWatch	Amazon's logbook. This is our proof that the job really ran.
Batch job	Work done in the background that nobody is waiting for. Exactly what we target.
ETL	A common batch job: moving data from one place to another and tidying it up.
Deadline	The time we promise the job will be finished by. We never go past it.
Dry run	Working out the answer and showing it, without actually booking anything.
Open source / MIT	The code is public and anyone may use it for free, including companies.

If you get asked something you do not know

Do not guess. Say this, calmly:

"That is the scheduler logic — Ramya built that part, she will take it."

That is honest, it takes two seconds, and it looks like a team that knows who owns what. Guessing wrongly looks far worse than handing it over.

Before you walk in: read this once more, look at the workflow diagram, and make sure you can say the one-sentence version in your own words. That is genuinely enough.